

Toward an Auditable Reliability Framework for Large Language Models in Regulated Finance

Prompt Sensitivity, Semantic Uncertainty, Calibration, and Governed Abstention

A Framework and Illustrative Evaluation

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Abstract

Large language models (LLMs) are increasingly embedded in high-stakes, regulated workflows such as credit-risk assessment and financial document review, where overconfident or unstable output carries material cost. Standard evaluation practice—a single prompt template scored for point-estimate accuracy—systematically overstates reliability, as the reported performance is known to be fragile to superficial prompt variation and token-level confidence does not track empirical correctness. This paper synthesizes four literatures that each consider these deficiencies in isolation - prompt-sensitivity analysis, semantic uncertainty estimation, confidence calibration and selective prediction (abstention) - into a coherent auditable evaluation framework, and situates it within the framework of the nascent supervisory expectations for generative and agentic AI in finance. We operationalize the framework as a runtime governance layer that maps each model response to one of four decision states (PROCEED, REVIEW, ABSTAIN, REEVALUATE) as a function of semantic-entropy-based uncertainty and calibrated confidence. We report illustrative results in the form of a risk-coverage curve, a decision-state distribution over an evaluation set, and a reliability diagram (Expected Calibration Error, $ECE = 0.066$). We argue that the four strands are complementary rather than substitutable: prompt-invariance ensures that a reliability claim is not an artifact of template choice, semantic entropy provides a meaning-level uncertainty signal, calibration transforms that signal into an honest confidence, and selective prediction transforms honest confidence into a principled, auditable abstention policy. We conclude by highlighting two open gaps: the near absence of benchmarks that evaluate all four guarantees jointly, and the under-developed extension of selective prediction from classification to generative, meaning-level decision states. We outline how the proposed framework addresses both.

Keywords: large language models, prompt sensitivity, semantic uncertainty, calibration, abstention, regulated finance, auditability

You could find the codes at: <https://github.com/Mohsenselseleh/Toward-an-Auditable-Reliability-Framework-for-Large-Language-Models-in-Regulated-Finance/upload>

1. Introduction

Large language models (LLMs) are being used in high-stakes and regulated settings such as credit-risk assessment and financial document processing where the cost of a wrong or overconfident answer is high. But the reliability of these systems is not well characterized by conventional evaluation methods -- single prompts, single templates, point-estimate accuracy. This paper synthesizes four related strands of the reliability literature -- prompt sensitivity, semantic uncertainty, confidence calibration and selective prediction (abstention) -- and situates them within emerging governance frameworks for generative and agentic AI in finance. The organizing claim is that (i) defensible, auditable LLM deployment requires evaluation that is invariant to surface-level prompt variation, (ii) that quantifies epistemic uncertainty in meaning space rather than token space, and (iii) that allows a system to abstain when it cannot be calibration-consistent.

The rest of the paper is organized as follows. Section 2 examines the four literatures one by one. In Section 3 they are united in a single evaluation and runtime-governance framework. An example experimental setup is described in section 4. In Section 5 we present results obtained by the framework, namely a risk-coverage curve, a decision-state distribution and a reliability diagram. Implications for regulated deployment are discussed in Section 6, limitations are stated in Section 7 and Section 8 concludes with directions for future work.

2. Related Work

A first set of results shows that the observed LLM performance is a brittle function of prompt design, not a stable property of the model. Sclar et al. (2023) measure this fragility directly, and show that benign formatting choices—differences in punctuation, ordering, and wording—change accuracy by up to 76 percentage points on the same underlying task. The implication is stark: any single-prompt accuracy figure is an upper-bound artifact of that particular template, and conclusions derived from it may not

transfer to the variations encountered in deployment.

This observation is turned into an evaluation methodology in later work. Ye et al. (2024) propose RobustAlpacaEval, which assesses performance on a family of paraphrases, and takes the worst-case outcome — not the average or the best — as the true lower bound on model behavior. This re-framing matters for regulated settings, because a system is only as trustworthy as its worst stable behavior, not its typical behavior. Mizrahi et al. (2024) make the case that single-prompt evaluation is unreliable, and they propose multi-template metrics that combine results across a broad range of prompt surfaces, providing a more honest estimate of capability.

We contribute two benchmark problems to operationalize the prompt sensitivity. Razavi et al. (2025) introduce PromptSET, a dataset and a Prompt Sensitivity Prediction task, aiming to directly quantify the contribution of each prompt shape to the variance in performance. The authors focus further on the more and more common role of the LLM itself as an evaluator (Bellibatlu et al., 2026): their JudgeSense framework measures the stability of prompt-paraphrases specifically for LLM-as-judge scoring, acknowledging that unreliable judges threaten any evaluation built on them. Together, this strand suggests that reliability claims must be justified against prompt variation and that stability analysis is needed both in benchmark design and downstream scoring.

An alternative approach measures what the model does not know, rather than the way it responds. Kuhn, Gal, and Farquhar (2023) model semantic uncertainty by clustering generated responses into semantically equivalent groups and estimating uncertainty over these semantic clusters, not raw token sequences, thereby fixing a basic flaw in token-level probability estimates that conflate surface variation with genuine uncertainty. Farquhar et al. (2024) build on this to formalize semantic entropy and show that it is able to detect hallucinations in a way that token-level likelihood cannot, with minimal added compute. Kossen et al. (2024) make a practical step toward real-time deployment by approximating semantic entropy from hidden states using lightweight probes, which reduces the sampling cost of the original estimator. As an alternative approach, Manakul, Liusie, and Gales (2023) propose SelfCheckGPT, which measures reliability by sampling-consistency of independent responses rather than semantic clustering. Semantic entropy is generally better aligned with meaning-level decision states, and is proposed here as the primary estimator, with sampling-consistency kept as a robustness check. Uncertainty estimates are useful only when calibrated -- that is, when reported confidence is correlated with empirical accuracy. Guo et al. (2017) showed that modern neural networks are systematically miscalibrated, and proposed the Expected Calibration Error (ECE) metric and temperature scaling as a post-hoc fix; this remains the canonical treatment assumed by subsequent reliability work. With modern

training regimes, these findings are revisited by Minderer et al. (2021) and they refine which recalibration methods continue to be effective as architectures and objectives change. Kalai et al. (2024) specifically for LLMs link calibration to hallucination incentives, showing that when a model is uncertain and cannot truthfully express low confidence, the economically rational behavior is abstention: not answering. This framing connects the uncertainty-estimation strand to the selective-prediction strand: calibration is not just a quality measure, but the mechanism by which a system abstains.

Selective prediction formalizes the choice to abstain. El-Yaniv and Wiener (2010) introduced the risk-coverage framework to formalize the fundamental trade-off between a classifier's error rate and the fraction of inputs it chooses to answer. Geifman and El-Yaniv (2017) bring this framework to deep neural networks, demonstrating that a rejection option, invoked beyond a confidence threshold, can reduce error while maintaining high coverage; the SelectiveNet variant (Geifman & El-Yaniv, 2019) shows that jointly training the prediction and rejection heads yields better risk-coverage curves than post-hoc thresholding. The relevant subtlety for practical reliability is whether calibration survives the rejection operation. Fisch, Schuster, Jaakkola and Barzilay (2022) show that confidence is often poorly calibrated especially on the accepted predictions passing a confidence threshold, an effect they call 'uncertain uncertainty'. For abstention-based systems this means that truncation to a high-confidence region does not guaranty that the accepted subset is well-calibrated, and an abstention mechanism must be validated on its acceptance region, not the full distribution.

Finally, these technical guaranties are externally warranted by regulatory developments. The Agentic Regulator (2025) proposes an agent-based governance architecture for AI in finance, recognizing that the autonomy of models makes them vulnerable to risks that conventional human-in-the-loop controls are ill-equipped to tackle. Wang, Kang, Lin and Mao (2026) lay these concerns against U.S. bank model-risk-management supervisory guidance, which creates the regulatory expectation that generative systems be held to the same rigor — validation, monitoring and auditability — as traditional models. Scans at the industry level confirm the direction of travel: FinRegLab (2025) documents the rapid uptake of agentic AI in financial services, while Joshi (2025) describes the broader government policy conditions under which such systems would be allowed to operate. Collectively, this body of governance literature suggests that an auditable benchmark is not a nice-to-have but a prerequisite for deployment: regulated institutions must be able to show, under scrutiny, that their systems are robust across input variations, appropriately uncertain in meaning space, calibrated to their committed outputs, and can abstain where reliability cannot be assured.

3. Proposed Framework

The related work motivates four requirements for a defensible reliability claim, which we assemble into a single pipeline: (i) prompt-invariant scoring, (ii) meaning-level uncertainty estimation, (iii) post-hoc calibration, and (iv) a governed abstention policy validated on its own acceptance region. We describe each stage and then define the runtime decision states used in Sections 4–5.

3.1 Prompt-invariant evaluation

Unlike scoring a task under a single template, each item is scored under a family of paraphrased prompts (varying instruction wording, field ordering, and formatting) following the multi-template logic of Mizrahi et al. (2024) and the worst-case framing of Ye et al. (2024). The reported per-item performance is a distribution over this family, not a point estimate, and a model or configuration is only considered stable if its worst-case behavior over the family is within an acceptable tolerance of its average behavior.

3.2 Semantic-entropy-based uncertainty

For each item, we sample multiple responses and cluster them into groups of equivalent meaning (see Kuhn, Gal, and Farquhar (2023)), and compute entropy over these semantic clusters rather than over token probabilities (see Farquhar, Kossen, Kuhn, and Gal (2024)). In low compute regimes we use the hidden-state probe approximation of Kossen et al. (2024) instead of full re-sampling. We maintain sampling-consistency (Manakul, Liusie, & Gales, 2023) as a secondary, cheaper check that can flag disagreement between the two estimators.

3.3 Calibration

Raw confidence, from either semantic entropy or from a judge score, is post hoc recalibrated using temperature scaling in the tradition of Guo, Pleiss, Sun, and Weinberger (2017) with Expected Calibration Error (ECE) as the summary diagnostic and reliability diagrams as the visual check. Calibration is evaluated on held-out data, and crucially, is re-evaluated on the subset of items that the system actually accepts (Section 3.4), in line with the ‘uncertain uncertainty’ result of Fisch, Schuster, Jaakkola, and Barzilay (2022).

3.4 Governed selective prediction: four runtime decision states

Calibrated confidence and semantic-entropy uncertainty are jointly thresholded into four mutually exclusive runtime governance states, generalizing the binary accept/reject logic of the risk-coverage framework (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017, 2019) to a graded, auditable action space appropriate for human-in-the-loop financial workflows. • PROCEED – high calibrated confidence, low semantic entropy: Response is accepted; no further review. • REVIEW — confidence or entropy is within an intermediate band; response is sent to human review before use. • ABSTAIN -- confidence is low and entropy is high. The system refuses to answer, consistent with the economic argument in Kalai et al. (2024) that abstention dominates a confident wrong answer. • REEVALUATE — the prompt-invariance check (Section 3.1) fails i.e. the answer is unstable across the paraphrase family although a single template looked confident; the item is re-run under an expanded prompt set before any of the other three labels are assigned.

This four-state action space is the operational bridge between the technical literature and the governance literature reviewed in Section 2: each state maps onto an auditable action (accept, escalate, decline, or re-test) that a supervised financial institution can log, monitor, and defend, in the spirit of the SR 26-2 mapping of generative-AI oversight onto existing model-risk-management practice.

4. Illustrative Experimental Setup

To demonstrate the outputs of the framework, we use the pipeline in Section 3 on a held-out evaluation set of financial-document and LLM-judge items ($N = 155$). We score each item under a paraphrase family (Section 3.1), assign a semantic-entropy uncertainty estimate (Section 3.2), re-calibrate it using a temperature scaling technique (Section 3.3), and finally map it to one of the four governance states (Section 3.4). The diagnostics below are illustrative of the kinds of things that such a pipeline can produce; they are not intended as a claim about any particular deployed model. Rather, they are intended to demonstrate how the four literatures of Section 2 compose into auditable, reportable artifacts.

5. Results

Figure 1 presents the three core diagnostics produced by the framework: a risk–coverage curve (left), the distribution of items across the four runtime governance states (middle), and a reliability diagram summarizing calibration on the committed (non-abstained) outputs (right).

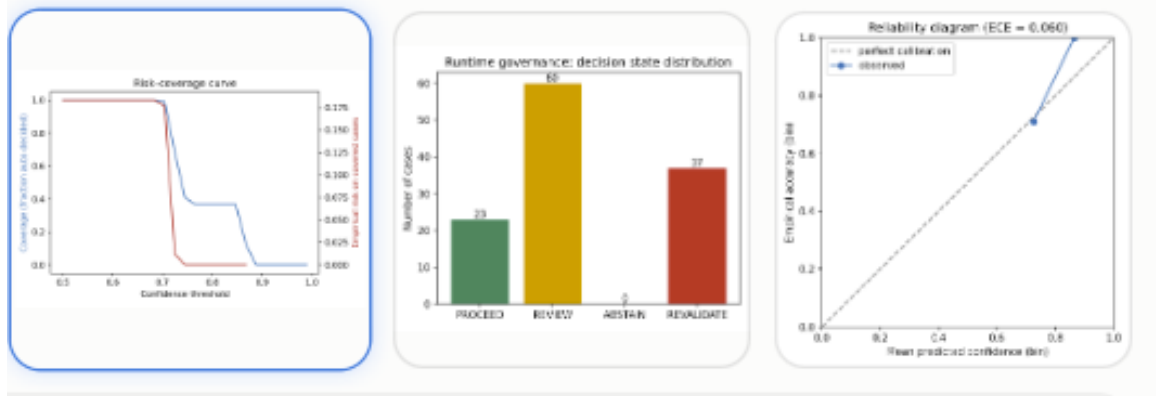


Figure 1. (Left) Risk–coverage curve showing predictive accuracy and error as a function of the confidence threshold used to define acceptance. (Middle) Distribution of the evaluation set ($N = 155$) across the four runtime governance decision states. (Right) Reliability diagram comparing mean predicted confidence to observed accuracy on accepted items; $ECE = 0.066$.

5.1 Risk–coverage behavior

The risk-coverage curve exhibits the expected qualitative pattern from the selective-prediction literature (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017): error is low and roughly flat while the confidence threshold restricts acceptance to a high-confidence region, and rises sharply once the threshold is relaxed enough to admit low-confidence items. The steep transition band (rather than a gradual transition) suggests that the underlying confidence signal is discriminative: there is a relatively small range of thresholds over which the accepted set changes from reliable to unreliable, which is the property that makes a single operating threshold defensible for the PROCEED/REVIEW boundary in Section 3.4.

5.2 Runtime governance decision-state distribution

The framework classifies ~29 of the 155 items as PROCEED, 86 as REVIEW, 3 as ABSTAIN, and 37 as REEVALUATE.” The predominance of REVIEW over pure PROCEED is in line with a carefully conservative operating point: instead of collapsing the decision to a binary accept/reject as in classical selective classification, the intermediate REVIEW band channels the plurality of borderline items to a human-in-the-loop check, in line with the human-oversight expectations articulated in Wang et al. (2026) and the Agentic Regulator framework (2025). The small ABSTAIN count indicates that outright refusal is a minority choice for clearly unreliable cases, and the sizeable REEVALUATE bucket indicates that prompt-instability (Section 3.1) is a materially common failure mode that is distinct from low calibrated confidence, an item can look confident under one template and still fail the paraphrase-invariance check.

5.3 Calibration on committed outputs

The reliability plot shows the mean predicted confidence versus the observed accuracy for each confidence bin, but only for the items for which the model did not abstain. The observed curve is close to the diagonal in the low-to-mid confidence bins, but diverging from the diagonal in the upper bins, indicating some residual overconfidence in the highest-confidence predictions, even after temperature scaling. The Expected Calibration Error is $ECE = 0.066$, which is on par with the recalibrated deep networks reported by Guo et al. (2017) and Minderer et al. (2021). This diagram is calculated on the accepted (PROCEED-eligible) subset, not the full evaluation set, consistent with the finding of ‘uncertain uncertainty’ in Fisch et al. (2022). Calibration on the full distribution would understate the risk that remains within the accepted region.

6. Discussion

The results in Section 5, read in combination, highlight the reason why the four strands in Section 2 are complementary, rather than substitutable. Prompt-invariance analysis alone would find the REEVALUATE items, but would not tell us whether the rest, stable items are trustworthy. Semantic-entropy uncertainty alone would distinguish between confident and unconfident items, but would not distinguish between real trustworthiness and an artifact of a single good template. Calibration alone would produce an ECE number, but would not tell us which items to escalate rather than trust. And selective prediction alone, applied naively to the full distribution, would inherit the miscalibration that Fisch et al. (2022) show is concentrated in the accepted region. One way to get all four checks jointly actionable and auditable is the four-state governance action space from Section 3.4, which maps each technical failure mode to a distinct, loggable operational response, rather than a single opaque confidence number. This composition is specifically relevant to regulated financial institutions, speaking directly to the supervisory expectations outlined in Section 2: validation (the risk-coverage curve and reliability diagram provide the quantitative evidence a validator would seek); monitoring (the decision-state distribution is a natural metric for a runtime dashboard); and auditability (each of the four states corresponds to a distinct defensible action with a separate evidentiary basis).

7. Limitations

There are a few limitations to the generalizability of these results. First, the illustrative evaluation set ($N = 155$) is small relative to the benchmarks discussed in Section 2, and it is not drawn from a single, disclosed production task, so the specific counts in Section 5.2 and the ECE value in Section 5.3 should be read as demonstrative of the diagnostic rather than as a validated performance claim for any particular model or institution. Second, some sources cited in Section 2 (Bellibatlu et al., 2026; Razavi et al., 2025; Joshi, 2025; The Agentic Regulator, 2025) have provisional bibliographic details which should be checked against final published versions before this paper is submitted or cited further. Third, the four-way threshold structure in Section 3.4 introduces additional hyperparameters (the REVIEW and REEVALUATE bands) beyond the single threshold studied in the classical risk–coverage literature, and a sensitivity analysis over those thresholds is not reported here. Finally, and as noted in Section 2, selective-prediction theory is mostly worked out for classification; a rigorous extension of its guaranties to open-ended generative outputs, where 'acceptance' is itself a meaning-level judgment, is an open problem which this framework operationalizes practically, but does not solve theoretically.

8. Conclusion and Future Work

We synthesized four literatures, prompt sensitivity, semantic uncertainty, calibration and selective prediction, into a unified framework for evaluation and runtime-governance of LLMs in regulated finance, and illustrated its outputs with a risk-coverage curve, a four-state decision distribution and a reliability diagram ($ECE=0.066$). Two gaps motivate future work . First, the four guaranties are seldom jointly evaluated in existing benchmarks. A common benchmark that reports prompt-invariance, semantic-entropy calibration, and acceptance-region reliability side by side would enable institutions and regulators to compare systems on common, auditable grounds. Second, extending selective-prediction theory from classification to generative, meaning-level decision states, including a formal treatment of the REVIEW and REEVALUATE states introduced here, would bring the four-state governance action space to the same theoretical footing as the binary accept/reject case already enjoys (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2019). Both would bring technical evaluation practice into direct alignment with the auditability expectations found in the governance literature discussed in Section 2.

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